



The Open University of Israel
Department of Mathematics and Computer Science

Thesis Proposal

Towards more Accurate ETA by Leveraging Future Traffic Predictions

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1 Introduction

Rapid urbanization and the growth of vehicle density have significantly increased the demand for efficient navigation systems. Current navigation tools, such as Google Maps and Waze, primarily rely on real-time traffic data to calculate routes and estimate travel times. Although effective, these systems are limited in their ability to anticipate future traffic conditions, leading to suboptimal routing decisions and inaccurate estimated time-of-arrival (ETA) predictions. For example, consider a scenario where a navigation system is aware of the destination requests and planned routes of multiple users in a city. Using this knowledge, the system can predict the future location of all vehicles and avoid potential congestion points and optimize routes for all users simultaneously. For example, if number of vehicles are predicted to intersect at a busy intersection at the same time, the system can suggest alternative routes that reduce the overall traffic load, resulting in shorter travel times for all vehicles. This holistic consideration of user paths and future locations represents a significant improvement over current systems that treat each user's route in isolation.

This thesis proposes to address these limitations by developing a dynamic navigation model that integrates destination requests and planned routes with real-time and historical traffic data to predict future vehicle location and hence future traffic conditions. By incorporating these predictions into a dynamic traffic graph, the model aims to provide more accurate ETAs and adapt routes dynamically to evolving traffic scenarios. The research will build directly on the foundational work described in [1]. To develop this advanced navigation model, the proposed approach will leverage a combination of techniques, including machine learning algorithms for traffic prediction, dynamic graph-based modeling of the road and vehicles network, and optimization methods for adaptive routing.

1.1 Problem Statement

Existing navigation systems excel at reacting to current traffic conditions, but cannot predict near-term traffic trends effectively. This gap leads to routes that can become suboptimal during the journey, causing delays and inefficiencies. The challenge lies in developing a scalable and computationally efficient model that can integrate predictive traffic data into real-time routing algorithms.

1.2 Research Question

How can a dynamic navigation model leverage user-defined destination requests and planned routes to predict congestion points and optimize ETA estimations and routing holistically across urban traffic networks?

2 Literature Review

The proposed research builds on existing work in traffic prediction and dynamic path-finding, with a primary focus on extending the concepts introduced by [1]. Their model integrates future vehicle positions into navigation systems to reduce ETA errors, predicting congestion points by estimating traffic loads based on vehicle distances and arrival times. While their approach demonstrated improved ETA accuracy by optimizing individual routes, it does not address the holistic optimization of multiple users' paths. This research aims to bridge that gap by developing a navigation model that simultaneously optimizes routes for all users, leveraging real-time data and future traffic predictions to enhance city-wide traffic efficiency.

Further, Wang et al. [2] highlighted the importance of incorporating spatial and temporal dependencies in traffic prediction models to achieve higher accuracy. Similarly, Liebig et al. [3] demonstrated the benefits of combining sensor data, historical trends, and discrete probabilistic models to enhance the accuracy of traffic flow predictions.

Dynamic Traffic Graphs Representing road networks as dynamic graphs with time-dependent edge weights has proven effective in modeling real-world traffic conditions. Examples include works by Claussen (1991) [4] on early navigation systems that accounted for real-time traffic information and Voloch et al. (2019) [5] on managing large distributed dynamic graphs for smart city networks .

Traffic Prediction Algorithms Machine learning models, such as regression and neural networks, have been used to forecast traffic density and travel times based on historical and real-time data. Notable contributions include Shi et al. (2020)[6], who explored predictive models for electric vehicle navigation, and Liu et al. (2015)[7], who integrated channel quality maps for load balancing in traffic networks. This line of research demonstrates the feasibility and potential benefits of incorporating traffic prediction into navigation systems, paving the way for the proposed work to further advance the state-of-the-art by developing a comprehensive model that can leverage both real-time and future traffic data to optimize routing and ETA estimates.

Pathfinding and Adaptive Routing Techniques Algorithms such as Dijkstra's [8] and A*[9] are widely used for calculating optimal routes in navigation systems. These algorithms have been adapted to handle dynamic edge weights, which are critical for modeling real-world traffic scenarios. However,

their scalability and efficiency in real-time applications often require further optimization. Advances in flow algorithms, such as those proposed by Edmonds and Karp (1972) [10] and Dinic (1970) [11], have inspired scalable solutions for dynamic pathfinding. Edmonds and Karp’s work on maximum flow problems introduced efficient techniques for optimizing network flows, while Dinic’s algorithm leveraged layered graph approaches for improved computational efficiency in large-scale systems.

These foundational methods have directly influenced real-world navigation systems. Dynamic routing applications, such as those employed in smart city initiatives, utilize scalable algorithms to manage congestion and dynamically update traffic conditions. For instance, Google Maps uses variations of Dijkstra’s and A* algorithms to compute the shortest path and optimize travel routes by incorporating real-time traffic data [12]. Similarly, Waze applies user-generated data and dynamic routing techniques to adaptively reroute vehicles based on traffic conditions [13].

Real-Time Feedback Mechanisms Voloch et al. (2019) [5] introduced approaches to update graph weights in response to real-time traffic conditions dynamically, a concept critical for ensuring accurate ETAs in evolving traffic scenarios.

 **Periodic and Spatial-Temporal Dependencies** Periodic dependencies refer to recurring traffic patterns observed over time, such as rush hours, weekend travel trends, or seasonal variations in road usage. These patterns can be identified using historical traffic data, enabling systems to anticipate congestion during predictable periods.

Spatial-temporal dependencies describe how traffic conditions at one location influence traffic in nearby areas over time. For instance, a high volume of vehicles on a highway during peak hours may lead to congestion not only on the highway itself but also on feeder roads and adjacent intersections as traffic spills over. Similarly, a major public event in a city center might cause delays that ripple outward, impacting roads several kilometers away as vehicles navigate alternative routes at different times.

Incorporating these dependencies into traffic prediction models, as demonstrated by Wang et al. [2], enhances the accuracy of forecasts and improves navigation decisions. A key novelty of this research is that future traffic includes the projected positions of vehicles, thanks to the system’s awareness of their routes and destinations. By leveraging this knowledge, the system can predict congestion points more accurately and adjust navigation recommendations proactively, offering a significant improvement over existing methods

that rely solely on real-time or historical data.

Multi-Modal Data Integration Liebig et al. [3] showcased the potential to leverage various data sources, including sensor data, historical trends, and discrete probabilistic models, to improve the accuracy of traffic flow predictions.

3 Proposed Methodology

This research focuses on developing a navigation model that integrates dynamic traffic graphs, predictive traffic modules, and feedback mechanisms to enhance routing accuracy and efficiency. The methodology involves the following components:

Dynamic Traffic Graph The dynamic traffic graph incorporates two types of nodes:

- **Static Nodes:** Represent intersections, points of interest, or road endpoints, capturing the physical structure of the road network.
- **Dynamic Nodes:** Represent vehicles, dynamically capturing their movement and interaction with the traffic graph.

The dynamic graph records the current positions of all vehicles in motion and incorporates their projected future positions, enabling it to represent predicted traffic conditions. This capability allows the system to proactively model evolving traffic scenarios, utilizing both real-time and forecast data to optimize routes effectively.

Predictive Traffic Module The predictive traffic module leverages machine learning to forecast future traffic conditions using both historical and real-time data. Techniques from the literature are utilized to incorporate:

- **Periodic Dependencies:** Identify recurring traffic patterns such as rush hours and seasonal trends.
- **Spatial-Temporal Dependencies:** Model how traffic conditions in one location influence nearby areas over time.
- **Multi-Modal Data Sources:** Integrate data from various sources, such as sensors, user reports, and historical trends, to enhance prediction accuracy.

Real-Time Feedback Mechanism A feedback mechanism continuously updates the dynamic traffic graph as new traffic information becomes available during a journey. The system dynamically re-optimizes routes by recalculating paths based on the updated traffic graph and predicted future conditions.

Adaptive Routing Algorithm The adaptive routing algorithm is designed to continuously re-plan routes, dynamically adjusting to predicted future traffic patterns. The algorithm prioritizes high accuracy and computational efficiency, enabling the system to handle evolving traffic conditions effectively.

Model Foundation and Enhancements The real-time navigation model depicted in [?] serves as the foundation for this work. This research enhances the predictive capabilities of the existing system by incorporating advanced forecasting techniques and optimizing the routing algorithm for improved efficiency and accuracy, ensuring its applicability in real-world navigation scenarios.

Data Collection The research leverages publicly available datasets to construct and validate the proposed navigation model:

- Use publicly available datasets, such as OpenStreetMap [14] for road networks and historical traffic data from open repositories.
- Simulate real-time data using tools like SUMO [15] or MATSim [16].

Simulation and Validation The proposed navigation model will be tested under simulated and real-world conditions:

- Test the model on a simulated urban traffic network with 50+ nodes and 150+ edges.
- Compare ETA accuracy and routing efficiency with baseline systems and existing benchmarks.
- Conduct a field study in a real-world urban environment to validate the performance of the proposed navigation system.

Performance Optimization To ensure scalability and adaptability, the research explores optimization techniques:

- Explore distributed computing methods to handle large-scale networks.
- Optimize graph update mechanisms for faster real-time recalculations.
- Enhance robustness to data quality issues, such as sensor failures or missing and corrupted information.

Overall, this research aims to develop a comprehensive navigation model that leverages future traffic predictions to provide accurate ETA estimates and adaptive routing in urban traffic networks, ultimately improving mobility and efficiency for drivers and city planners.

4 Proposed Dynamic ETA Optimization Algorithm with Traffic Feedback

Accurate ETA estimation and efficient routing in urban traffic networks require dynamic adaptation to real-time and predicted future traffic conditions. This section presents the proposed algorithm, which integrates real-time and predictive traffic data into a dynamic graph framework. The algorithm optimizes routes by leveraging a machine learning-based predictive module, continuously updated traffic conditions, and dynamic recalculations to ensure high accuracy and adaptability. Below, we outline the input parameters, procedure, and outputs of the proposed algorithm:

1. Input

- **Static Graph $G = (V, E)$:**
 - V : Static nodes representing intersections, points of interest, or road endpoints.
 - E : Edges representing roads or lanes with static attributes (e.g., length, speed limits).
- **Dynamic Graph $G'(V', E', t)$:**
 - $V' = V \cup V_c$: Includes static nodes V and dynamic nodes V_c representing vehicles.
 - $E' = E \cup E_c$: Includes edges E from G and edges E_c representing vehicle connections and dynamic paths.
 - t : Snapshot of the traffic graph at a specific moment in time, where $t = 0$ represents the current time and $G'(V', E', t = 0)$ reflects the current traffic conditions. For example, $G'(V', E', t = 5)$ reflects the predicted traffic conditions 5 time units after $t = 0$.
 - $w'(e', t)$: Weight function representing travel time for edge e' based on the snapshot at time t , calculated using:
 - * Real-time traffic conditions.
 - * Predicted future traffic from periodic and spatial-temporal dependencies.
 - * Planned routes of vehicles.
- s : Source node (starting point).
- d : Destination node (target point).

2. Procedure

1. Step 1: Find the Current Best Route from G' at $t = 0$:

- At $t = 0$, compute the shortest path in $G'(V', E', t)$ from s to d , based on the weights $w'(e', t)$.
- Use a path-finding algorithm (e.g., Dijkstra's or A*) to identify the current best route considering real-time traffic at $t = 0$.

2. Step 2: Define t_d and $t_{d_{\max}}$:

- Calculate t_d , the total duration required to reach node d from node s along the selected path:

$$t_d = \sum_{e' \in p} w'(e', t)$$

- Define $t_{d_{\max}}$, the adjusted duration, as:

$$t_{d_{\max}} = t_d \cdot t_f$$

where t_f is a factor representing periodic and spatial-temporal traffic conditions, accounting for expected variations in traffic flow.

3. Step 3: Create the Set of Subgraphs G_s :

- Prune $G'(V', E', t)$ at each time step t_i from t_0 to $t_{d_{\max}}$ to create a subgraph $G_s^{t_i}$.
- Include in $G_s^{t_i}$ only the nodes and edges that can provide alternative routes to node d . This involves:
 - Removing nodes and edges that cannot be part of any path leading to d .
 - Ensuring that the subgraph maintains connectivity between s , d , and viable alternative routes.
- Generate a set of subgraphs $G_s = \{G_s^{t_0}, G_s^{t_1}, \dots, G_s^{t_{d_{\max}}}\}$.

4. Step 4: Use ML to Compute the Optimal Route p_{opt} :

- Feed G_s , s , and d into a trained machine learning model designed to evaluate the subgraphs and identify the shortest route from s to d .
- The ML model takes as input:

- Subgraph features, including nodes, edges, weights $w'(e', t_i)$, and connectivity.
- Source and destination nodes.
- The output is p_{opt} , the optimal route, based on the model’s learned predictions, incorporating the sub-graphs’ temporal and spatial variations.

5. Step 5: Update Traffic Conditions in G' :

- Add the vehicle position as a dynamic node v_{vehicle} to G' at time t , reflecting its location along p_{opt} based on its current progress.
- Adjust the weights $w'(e', t)$ of edges connected to v_{vehicle} , accounting for the impact of the vehicle’s movement on traffic flow.
- Update neighboring edges to incorporate potential congestion effects caused by the vehicle’s presence.

6. Step 6: Dynamic Recalculation (if necessary):

- During travel, monitor real-time updates and recompute the optimal path if significant changes occur in $w'(e', t)$ or $G'(V', E', t)$.

3. Output

- **Optimal Route:** The path p_{opt} from s to d with the minimum travel time, considering future traffic at time t .
- **Updated Traffic Graph:** Adjusted $G'(V', E', t)$ reflecting the addition of vehicle positions and recalibrated weights $w'(e', t)$.

5 Expected Outcomes

The proposed research is expected to deliver the following outcomes:

- A dynamic navigation model capable of integrating predictive traffic data.
- Improved ETA accuracy compared to baseline systems.
- Real-time adaptability to traffic changes with computational efficiency.

Task	Duration
Literature Review	2 months
Data Collection and Simulation	3 months
Model Development	4 months
Testing and Validation	3 months
Optimization and Refinement	3 months
Thesis Writing	3 months

Table 1: Research Timeline

6 Timeline

As shown in Table 1, the proposed research is expected to be completed within 18 months.

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